

Calibrated segmentation of CBCT and CT images for digitization of dental prostheses

Veerle Wouters · Wouter Mollemans · Filip Schutyser

For CBCT threshold of the guide

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Abstract

Purpose State of the art computer aided implant planning procedures typically use a surgical template to transfer the digital 3D planning to the operating room. This surgical template can be generated based on an acrylic copy of the patient's removable prosthesis—the so-called radiographic guide—which is digitized using a CBCT or CT scanner. Since the same accurate fit between the surgical template and the patient as with the radiographic guide and the patient should be ensured, a procedure to accurately digitize this guide is needed.

Methods A procedure is created to accurately digitize radiographic guides based on a calibrated segmentation. Therefore, two steps have to be executed. First, during a calibration step a calibration object is CBCT or CT scanned and a calibration algorithm which results in an optimal threshold value is executed. Next the guide is CBCT or CT scanned and a 3D model is created using the obtained optimal threshold. To validate our method, we compared a high accuracy laser scanned copy of the guide with the generated 3D model by creating a distance map between both models.

Results The procedure was performed for different CBCT and CT scanners, and the digitization error for each scanner was defined. The 90th percentile error measured on average 0.15 mm, which was always less than the applied voxel size for all CBCT and CT test scans.

Conclusions The calibration procedure evaluated in this study solves the known problem of digitizing a radiographic guide based on non-standardized gray value CBCT images. The procedure can easily be executed by a clinician and allows an accurate digitization of a radiographic guide using

a CBCT or CT scanner. Starting from this digitization, an accurate surgical template can be made which has a good fit on the patient's remaining teeth and surrounding soft tissues.

Keywords Three-dimensional imaging · 3D implant planning · Segmentation

Introduction

Contemporary oral rehabilitation treatments based on implants target a functional, esthetic, stable and reliable outcome with minimal pain, and short treatment times. To meet these criteria, for both partial edentulous and edentulous patients, 3D preoperative planning and adequate tools to transfer the planning to the operating room are valuable.

Verstreken et al. [1] introduced a double scanning procedure in which two scans are made, one of the patient wearing a prosthesis and one of the prosthesis separately. Currently, this method is seen as a standard of good clinical practice for “3D prosthetic driven” planning. The double scanning procedure enables to create a 3D model of the bone and the prosthesis and to align them with each other (Fig. 1a). Based on these inputs, a 3D implant planning can be made (Fig. 1b).

To transfer this 3D implant planning to the operating room, a surgical template that is generated based on the patient's prosthesis and the planned position of the implants can be used [2–13]. This surgical template contains tubes to guide the surgeon to the planned location of the implants (Fig. 1c). To correctly transfer the planning to the operating room it is very important that the fit between the surgical template on one hand and the soft tissue and the remaining teeth on the other hand is correct. Since the surgical template is generated from the scanned prosthesis, the digitization of this prosthesis should be as accurate as possible. Instead of a prosthesis,

V. Wouters (✉) · W. Mollemans · F. Schutyser
Nobel Biocare c/o Medicim NV, Kardinaal Mercierplein 1,
2800 Mechelen, Belgium
e-mail: veerle.wouters@medicim.com

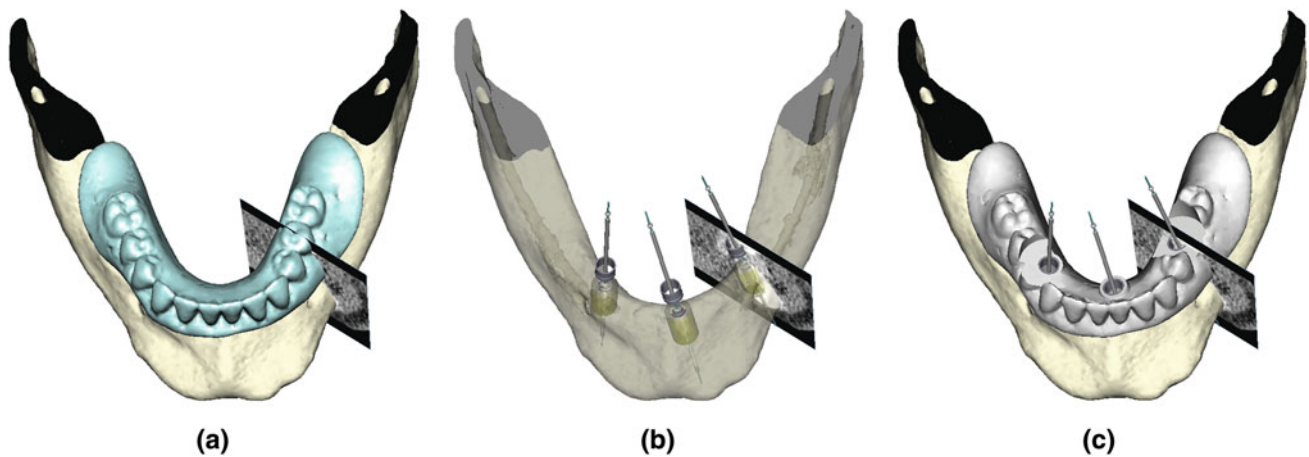


Fig. 1 **a** 3D model of the bone and the radiographic guide, **b** 3D model of the bone and implants at their planned locations, **c** 3D model of the bone, the planned implants and the surgical template, containing tubes at the position of the planned implants

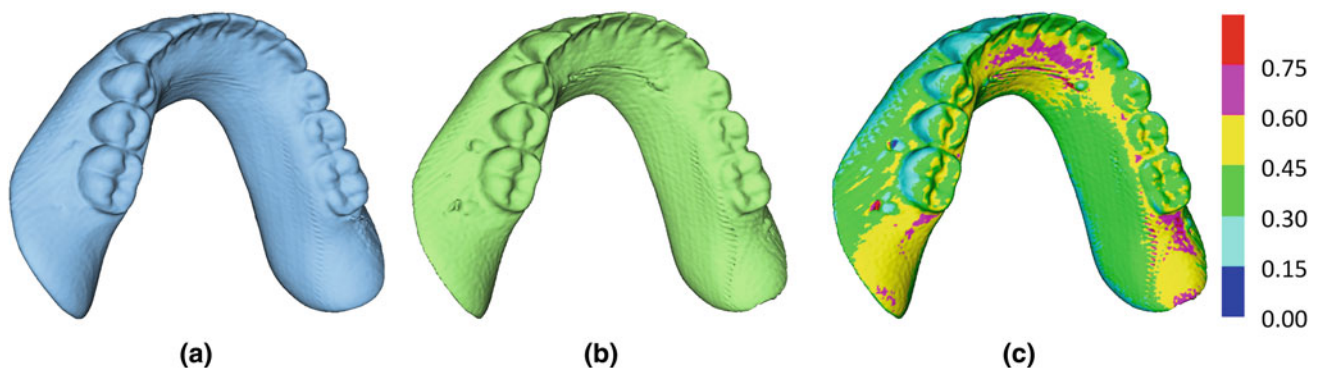


Fig. 2 3D model of radiographic guide after digitization with threshold **a** –700 HU and **b** 0 HU. **c** A distance map between both 3D models (mm)

an acrylic copy of an optimized prosthesis is typically used. This is the so-called radiographic guide.

To extract a digital 3D model of the guide out of a cone beam (CB) CT or CT scan, the marching cubes algorithm can be used [14]. This algorithm uses a threshold value to define the transition between the material and the background. To correctly digitize the patient's bone and the radiographic guide, a correct threshold has to be defined. In Fig. 2a and 2b, a guide is shown which is digitized with a threshold of respectively –700 Hounsfield Units (HU) and 0 HU. A distance map between both models (Fig. 2c) clearly shows that there is a significant difference between both 3D models. This example demonstrates the importance of selecting a correct threshold in order to obtain an accurately digitized guide and consequently to ensure a precise fit between the patient and surgical template during surgery. The purpose of this paper is to develop a procedure to easily define an optimal threshold needed to reconstruct a CBCT or CT scanned dental guide as accurate as possible.

Two types of CT scanners exist: medical CT and CBCT. A medical CT uses calibrated HU values. This means that the gray values are always the same for a certain type of mate-

rial. For these scanners, the calibration procedure should only be executed once to know the optimal threshold for a certain material scanned with any type of medical CT scanner. CBCT scanners, however, do not have such calibrated HU values. The gray values can differ for each scanner. Therefore, an easily repeatable procedure is necessary defining the gray value of a certain material in a certain scanner. It should be possible for clinicians to do this procedure themselves and they should be able to redo it with a regular frequency in time, or when changes or updates to the CBCT scanning equipment occur. This was taken into account when designing the proposed procedure.

Material and methods

The developed procedure consists of two steps: a calibration step and a calibrated segmentation step. In short, the calibration step works as follows. The procedure starts from a known shape called the calibration object. This object is scanned with a CBCT or CT scanner and the resulting scan is compared with the ground truth 3D surface using a calibration algorithm. The outcome of this calibration algorithm is

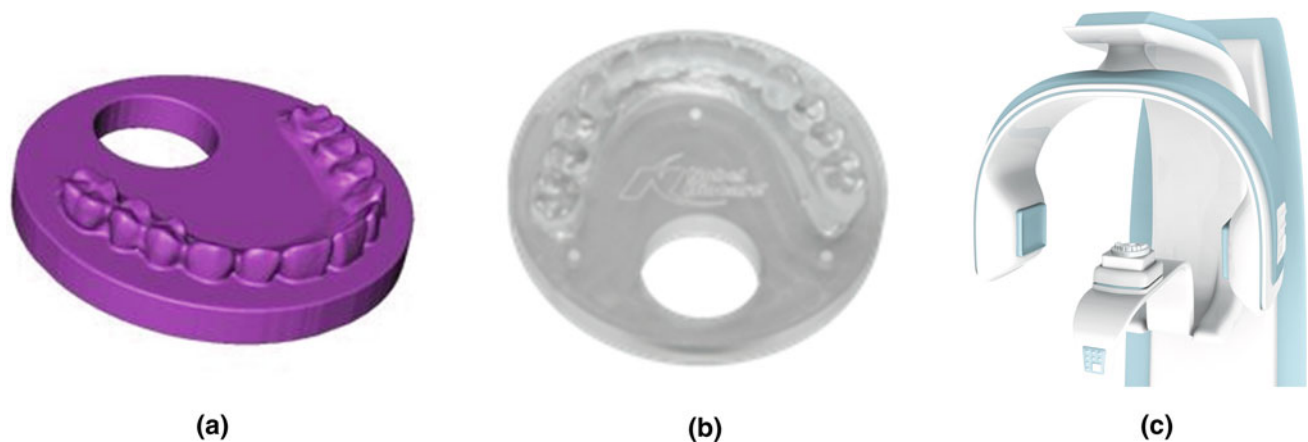


Fig. 3 **a** A virtual calibration object, **b** a fabricated calibration object, **c** the calibration object positioned in a scanner at the same location as the radiographic guide

an “optimal threshold” for which the estimated digitization error is minimal. This optimal threshold is afterward used in the calibrated segmentation step in which a 3D surface model of a scanned radiographic guide is computed.

The calibration step should be executed once for medical CT scanners or with a regular frequency for CBCT scanners, while the calibrated segmentation step is performed each time a radiographic guide is digitized from CBCT or CT images.

Design of calibration object

For the design of the calibration object, some specific guidelines are considered. First, the volume of the designed object is preferably more or less equal to the typical volume of a dental prosthesis. Moreover, it is preferred that the surface of the object contains sufficient detailed 3D information, i.e. shape variation, so that the accuracy of the algorithm can be guaranteed. Finally, the properties of the material used for the calibration object should be similar or equal to those of the radiographic guide.

Keeping these guidelines in mind, a calibration object was designed consisting of a typical dental surface virtually mounted on a cylinder with a small height (Fig. 3). The material of the object is similar to the material typically used to fabricate radiographic guides. A 3D image volume and the exact shape information of this calibration object are known.

Calibration algorithm

During the calibration procedure, the calibration object is scanned with a CBCT or CT scanner, which results in a 3D image volume of the calibration object (Fig. 4 (2)). Based on the image data from this scan and the known shape from the design, the optimal threshold to generate the shape of the calibration object is determined. In addition to this, the esti-

mated digitization error can be computed. Figure 4 shows the complete outline of the algorithm. In the remainder of this section, we discuss this algorithm in more detail. The algorithm consists of three major steps: aligning the two input data sets (a), building a histogram (b), and finally deriving the optimal threshold value (c).

We start from the calibration object design, i.e. the ground truth model, and the image volume of the CBCT or CT scanned calibration object. Since we have to compare both data sets to derive the optimal threshold, both models should first be aligned (Fig. 4 (4)). This alignment is obtained by voxel-based rigid registration using the maximization of mutual information (MI) criterion [15] of the ground truth image volume and the image volume of the CBCT or CT scanned calibration object. As known, the MI registration needs a good initialization. We automatically derive this initialization by aligning the main directions of the scanned calibration object to the known main directions of the ground truth surface. To find the main directions of the scanned object, the scanned image data are first binary segmented using the Otsu threshold method [16]. Next, a singular value decomposition is calculated on this binary segmented image, returning the main directions of the scanned object. After setting this initialization and executing the MI registration, the resulting transformation is applied to the ground truth 3D surface to obtain an aligned ground truth surface (Fig. 4 (5)). This aligned ground truth surface shares the same 3D space with the image volume of the scanned calibration object.

In a second step, the algorithm measures the gray values in the image volume of the calibration object at the surface of the aligned ground truth (Fig. 4 (6)). All measured gray values are stored and a histogram is built (Fig. 4 (7)). To improve the stability of the algorithm, the measure area is extended toward a small region around the surface

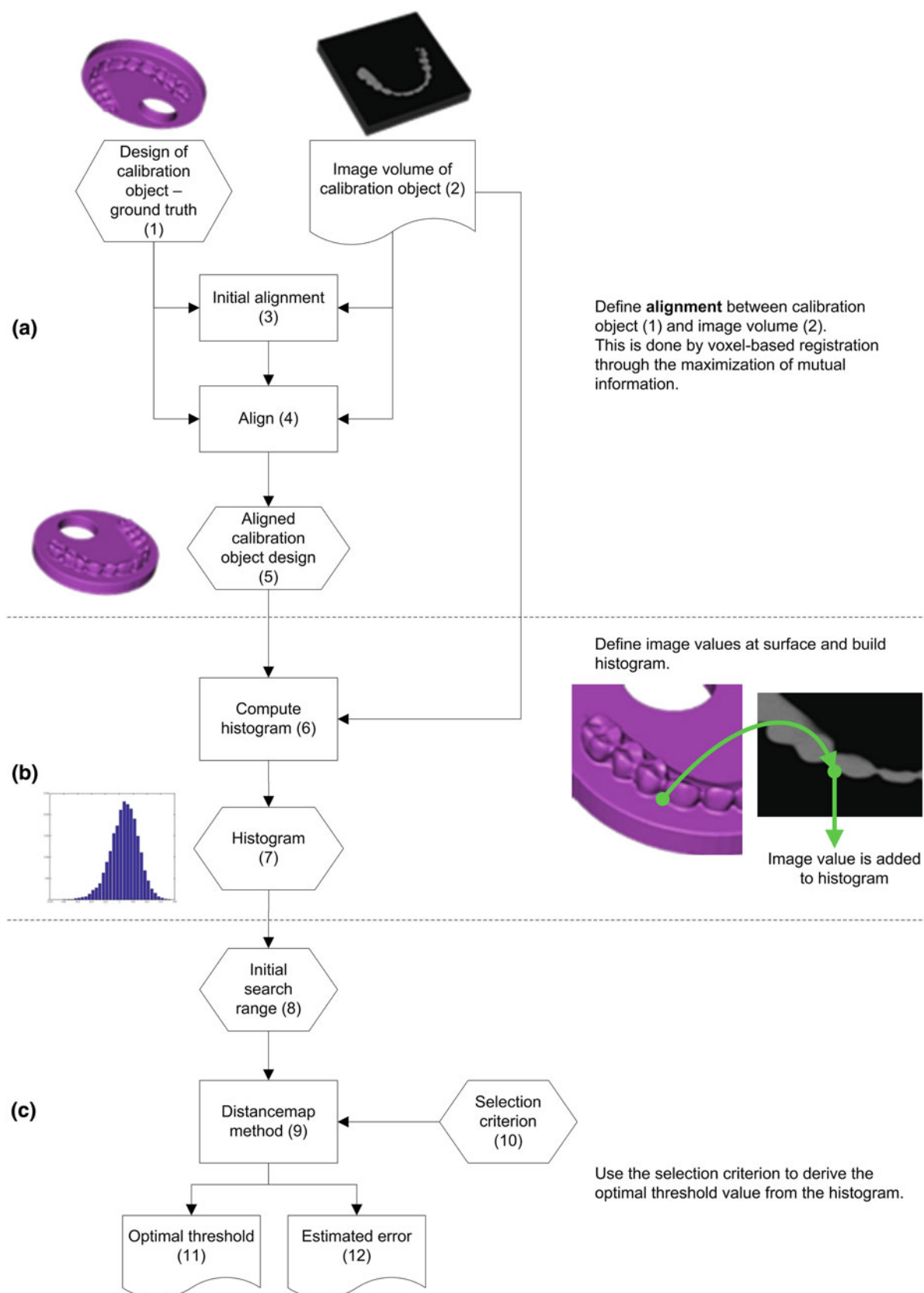


Fig. 4 Overview of the complete calibration algorithm containing three major steps: **a** aligning two input data sets, **b** building a histogram and **c** deriving an optimal threshold

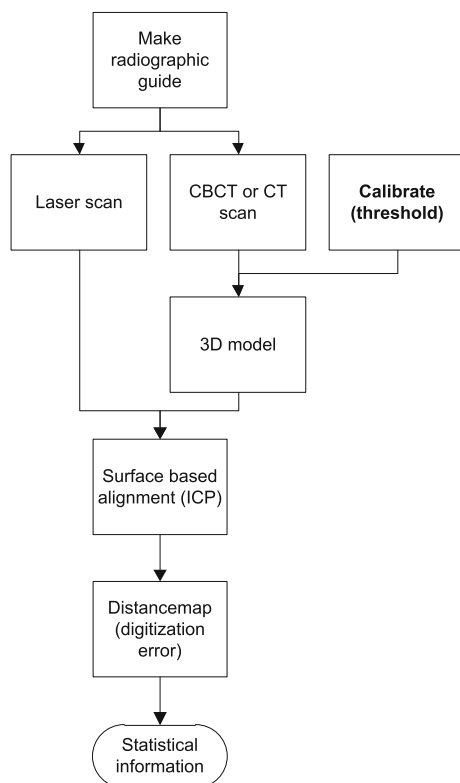


Fig. 5 Validation workflow comparing the laser scanned guide with the guide extracted out of a CBCT or CT image with the calibrated segmentation method

of the aligned ground truth. In this way, small errors during alignment or noise in the scanned data can be partially eliminated.

In the final step (c), the actual optimal threshold value is searched using an iterative refinement technique. First an initial search range for the optimal threshold value is defined starting from the generated histogram. This initial range includes all gray values which have at least half the height of the peak of the histogram (Fig. 4 (8)). Afterward in each iteration, a step size is defined and the current search range is sampled using this step size. As such a collection of optimal threshold candidates is defined. For each candidate, a surface representation is generated out of the scanned image volume using the marching cubes algorithm [14] and a distance map between this surface representation and the ground truth surface is calculated (Fig. 4 (9)). The 90th percentile of each distance map is calculated and defined as the estimated digitization error (Fig. 4 (12)). The candidate which leads to the smallest estimated error is selected and a new narrower search range around this value is defined measuring half the size of the previous search range. This process is repeated until convergence leading to the gray value with the overall smallest estimated error called the “optimal threshold” (Fig. 4 (11)).

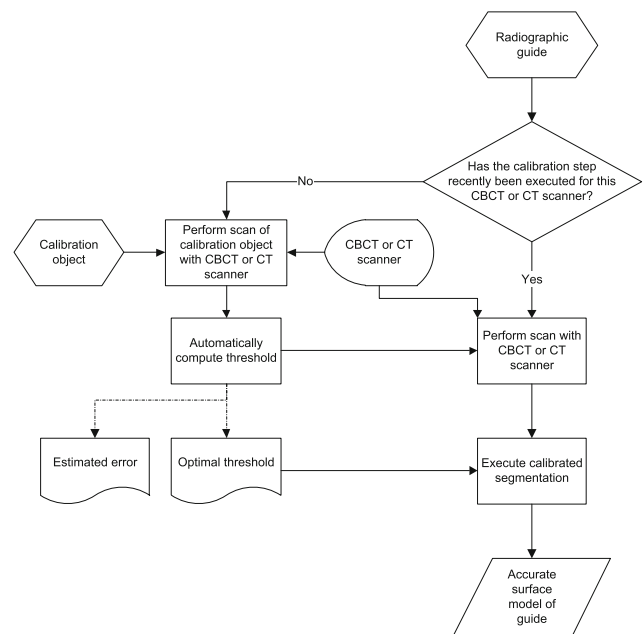


Fig. 6 Overview of the calibration procedure used for accurately digitizing a radiographic guide as introduced in the daily clinical workflow

Validation of the accuracy of the digitized radiographic guide

When generating a surgical template, two error sources can be defined. First a digitization error might arise when digitizing the radiographic guide from the CBCT or CT image data. Next a fabrication error will be introduced when producing the surgical template using a rapid prototyping technique (e.g. stereolithography). In this work, we only validated the digitization error, since our goal is to minimize this error. The validation was performed as described below.

First a guide was scanned with a CBCT or CT scanner. Using the optimal threshold obtained with the calibration algorithm, a 3D surface was made of the guide. To compare this 3D surface model with the original guide, a high accuracy optical scan was acquired from the original guide resulting in a scanned surface model. Both models were surface-based aligned using the iterative closest point (ICP) algorithm [17,18]. A distance map between the aligned models was calculated and statistically analyzed. An overview of the different steps is shown in Fig. 5.

Clinician setting

Figure 6 gives an overview of the calibration procedure used for accurately digitizing a radiographic guide as introduced in the daily clinical workflow. If the calibration step has not been executed recently, this step should be performed. During this step, a scan of the calibration object is made with the same protocol as the one which will be used to scan the

radiographic guide. Afterward the optimal threshold is automatically calculated together with the estimated digitization error. This optimal threshold is used to execute a calibrated segmentation of the guide resulting in an accurate surface model of the guide. When the calibration step was recently executed, the calibration object does not have to be scanned. Only the scan of the guide and the calibrated segmentation have to be performed.

Results

Calibration object

To produce the calibration object, a milling technique was used. A medical approved (USP Class VI) polycarbonate was selected as ground material. This material has similar radio-opacity as a radiographic guide.

Calibration algorithm

The calibration procedure was successfully executed for different scanners. The calibration algorithm was executed on a Windows Vista computer (Intel Core Duo T2400 CPU, 1.83 GHz, 2 GB RAM) and took an average time of 11 min. The histogram of the population of the gray values at the surface of the aligned ground truth surface (Fig. 4 (7)) is presented in Fig. 7 for a Siemens Sensation 64 CT scanner with a pitch of 0.9, an effective tube current of 90 mAs and a slice increment of 0.3 mm. The optimal threshold and estimated error for the different scanners are shown in Table 1. The optimal threshold was located around -500 for all scanners except for the E-Woo Picasso Trio CBCT scanner for which

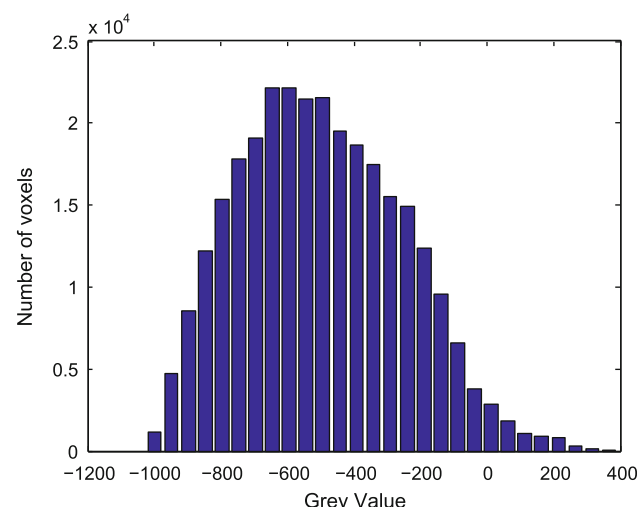


Fig. 7 Histogram of the grey values at the air-material interface of the calibration object for a Siemens Sensation 64 scanner

Table 1 The optimal threshold and estimated error (in mm) for different scanners

Scanner	Voxel size	Optimal threshold	Estimated error
Next Gen i-CAT	0.2	-448	0.21
Next Gen i-CAT	0.4	-416	0.25
E-Woo Picasso Trio	0.2	-1,012	0.22
GE Discovery CT750 HD	0.3	-548	0.22
3M Imtek Iluma	0.2	-494	0.18
Soredex Scanora 3D	0.2	-659	0.22
Siemens Sensation 64 (H30s)	0.3	-459	0.24
Siemens Sensation 64 (H70h)	0.3	-526	0.19
Newtom 3G	0.2	-570	0.21

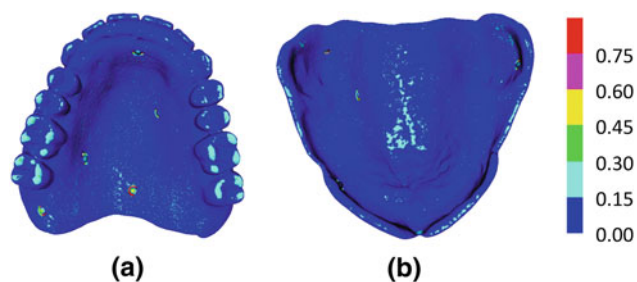


Fig. 8 Distance map between the radiographic guide and the digitized 3D model (in mm) for a Siemens Sensation 64 scanner

the threshold lies around -1000 . When comparing the estimated error for classical i-CAT with a voxel size of 0.2 mm and 0.4 mm, we see that the estimated error is smaller when using a smaller voxel size. As shown for the Siemens Sensation 64 scanner, setting a specific reconstruction filter can also influence the estimated digitization error.

Validation

An optical scan of the radiographic guide was made with ATOS IIe (GOM, Braunschweig, Germany). This optical scanner has a known measuring accuracy of 0.01 mm. Figure 8 shows the obtained distance map between the optically scanned guide and the reconstructed 3D model for the before-mentioned Siemens CT scanner by means of a color code. The corresponding distance distribution is shown in Fig. 9. In Table 2, the digitization errors for some other scanners are shown. As can be seen the average 90th percentile measures only 0.15 mm which is less than the applied voxel size for all test scans.

Discussion

In this work, we present a procedure to digitize a radiographic guide using a CBCT or CT scan, as accurate as possible. This

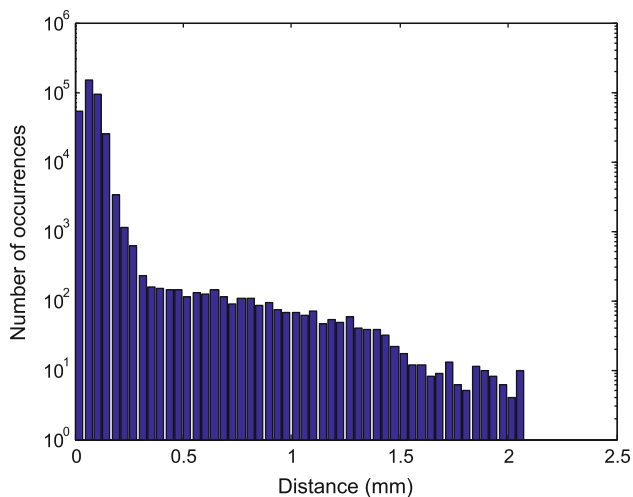


Fig. 9 Histogram of the distances (in mm) between the laser scanned radiographic guide and the 3D model digitized with the calibrated segmentation method for a Siemens Sensation 64 scanner

Table 2 The digitization error for different scanners (in mm). P50 = 50th percentile, P90 = 90th percentile, P95 = 95th percentile and P99 = 99th percentile

Scanner	Voxel size (mm)	P50	P90	P95	P99
Next Gen i-CAT	0.2	0.09	0.14	0.16	0.23
E-Woo Picasso Trio	0.2	0.11	0.19	0.23	0.35
GE Discovery CT750 HD	0.3	0.09	0.14	0.16	0.20
3M Imtek Iluma	0.2	0.07	0.11	0.13	0.15
Soredex Scanora 3D	0.2	0.10	0.18	0.20	0.24
Siemens Sensation 64 (H30s)	0.3	0.11	0.18	0.20	0.28
Siemens Sensation 64 (H70h)	0.3	0.08	0.13	0.15	0.23

procedure is more essential for a CBCT scan than for a CT scan since CBCT images do not use calibrated HU values. Unlike other studies [19,20], this study does not compare the quality and artifacts of CBCT scanners with CT scanners.

The design of the calibration object was done so that the different guidelines were followed and that the object has similar material properties as a radiographic guide. The proposed procedure can moreover be executed easily by a clinician or one of his team members in a short time. This easy setup enables execution of the calibration step in daily clinical practice on a regular frequency in time, or when changes or updates to the CT-scanning equipment occur.

Our results show that the usage of the proposed calibration procedure ensures a high accurate digitization of the CBCT or CT scanned radiographic guide. For all test scans, the 90th percentile of the digitization error measured less than 0.2 mm, which was in all cases less than the applied voxel size. Moreover, there was a clear correlation between the estimated digitization error, which is defined during calibration, and the actual measured digitization error. Therefore,

we conclude that this estimated error offers a good indication of the accuracy of a specific scanner with a specific protocol for scanning dental prostheses. Finally, it can be deduced from our results that setting a smaller voxel size or selecting a sharper reconstruction filter can improve the digitization accuracy. This conclusion is in line with what was expected.

During our tests various problems with specific CBCT scanning devices or protocols were detected. Some examples are: detection of gray value inhomogeneities, overexposure artifacts... The detection of these problems emphasizes the generality and usefulness of our method, not only for accurate digitization of dental prostheses. We therefore suggest that the proposed method (or a derivative approach) could be applied by clinicians for each new scanner or scanning protocol before this scanner or protocol is introduced in the clinical practice to guarantee accuracy and detect possible failures.

At last, the proposed procedure to accurately digitize a radiographic guide can also be used for digitization of other dental objects. However, it has to be noted that the to-be-digitized dental object should be homogeneous and that the calibration object should be adapted so that the material, volume, and shape of this calibration object matches the properties of the to-be-digitized dental object.

Conclusion

In this work, we presented a complete procedure to accurately digitize a dental model, e.g., a radiographic guide, which includes a calibration step and a calibrated segmentation step. The method was validated for different CT and CBCT scanners and successfully introduced into clinical practice. As discussed, the application field of the proposed procedure is in our opinion more general than only the digitization of a radiographic guide.

Conflict of interest All authors are employees of Nobel Biocare.

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